Evolutionary Conservation and Functional Constraint of TP53 Mutation Hotspots Across Mammalian Species

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ABSTRACT

The tumour suppressor protein TP53 is essential for preserving genomic integrity and is among the most frequently mutated genes in human cancers. Mutations in TP53 show a non-uniform distribution and are concentrated at specific residues, commonly referred to as mutation hotspots. Although these positions are well characterised in clinical studies, their evolutionary behaviour across species remains insufficiently understood.

In this study, TP53 protein sequences from ten mammalian species were analysed to investigate conservation patterns at six canonical hotspot residues (R175, G245, R248, R249, R273, and R282). Multiple sequence alignment was performed, followed by residue-level conservation scoring and statistical comparison with non-hotspot positions.

Hotspot residues exhibited significantly higher conservation (mean ≈ 0.95) compared to non-hotspot residues (mean ≈ 0.78), with the difference confirmed by the Mann–Whitney U test ($p = 0.034$). In addition, hotspot positions showed reduced variability, indicating stronger evolutionary constraint. Sequence identity analysis revealed moderate to high similarity (69–92%) across species, supporting the robustness of the dataset. Structural and alignment-based observations further demonstrated that hotspot residues are localised within a highly conserved functional domain.

These findings suggest that TP53 mutation hotspots represent evolutionarily constrained residues embedded within critical structural regions, providing insight into their susceptibility to recurrent mutation in cancer.

Keywords
TP53; mutation hotspots; evolutionary constraint; comparative genomics; protein conservation; cancer biology

1. INTRODUCTION

The TP53 gene encodes a transcription factor that functions as a central regulator of cellular responses to genomic stress, including DNA repair, apoptosis, and cell cycle arrest [1,2]. Due to its critical role in maintaining genomic stability, TP53 is among the most frequently mutated genes in human cancers [3,4].

Notably, mutations in TP53 are show a non-uniform distribution and are concentrated at specific residues, commonly referred to as mutation hotspots [5,6]. These hotspots are predominantly located within the DNA-binding domain, a region essential for transcriptional regulation and protein functionality.

These observations indicate that mutation hotspots are likely driven by functional vulnerability within structurally constrained regions [7]. However, it remains unclear whether TP53 hotspot residues are uniquely conserved compared to other residues within functionally constrained regions.

Comparative sequence analysis across species provides an effective framework for identifying conserved and variable regions within proteins. By analysing evolutionary patterns, it is possible to infer functional importance and structural constraints.

In this study, TP53 sequences from ten mammalian species were analysed using multiple sequence alignment and statistical methods to evaluate conservation patterns at hotspot residues. The objective was to determine whether these positions exhibit distinct evolutionary characteristics compared to non-hotspot residues.

2. MATERIALS AND METHODS

2.1 Sequence Retrieval

TP53 protein sequences were obtained from publicly available databases, including UniProt and NCBI Protein [1]. Only complete and high-confidence sequences were included, while partial or low-quality entries were excluded to ensure dataset reliability.

2.2 Multiple Sequence Alignment

Multiple sequence alignment was performed using Clustal Omega with default parameters [2,3]. The alignment ensured consistent positional comparison across homologous residues.

2.3 Conservation Analysis

Residue-level conservation scores were calculated based on the frequency of the most common amino acid at each alignment position.

2.4.Hotspot Residue Selection

Six canonical TP53 hotspot residues (R175, G245, R248, R249, R273, and R282) were selected based on established cancer mutation studies [4-6].

2.5.Statistical Analysis

Differences in conservation between hotspot and non-hotspot residues were evaluated using the Mann–Whitney U test.

2.6.Sequence Identity Analysis

Pairwise sequence identity was assessed to evaluate evolutionary divergence among species[7].

2.7. Phylogenetic Analysis

Phylogenetic relationships were inferred using standard evolutionary approaches [8,9].

2.8 Computational Tools

All analyses were performed using Python libraries including Biopython, NumPy, SciPy, and Matplotlib.

3. RESULTS

3.1.Conservation of TP53 Hotspot Residues

Conservation analysis revealed that TP53 hotspot residues are highly conserved across mammalian species, with scores ranging from approximately 0.91 to 1.00 and a mean value of 0.95. In contrast, non-hotspot residues exhibited greater variability, with a mean conservation score of approximately 0.78.

3.2.Statistical Comparison

Statistical analysis demonstrated a significant difference between hotspot and non-hotspot conservation scores ($p = 0.034$), indicating stronger evolutionary constraint at hotspot positions.

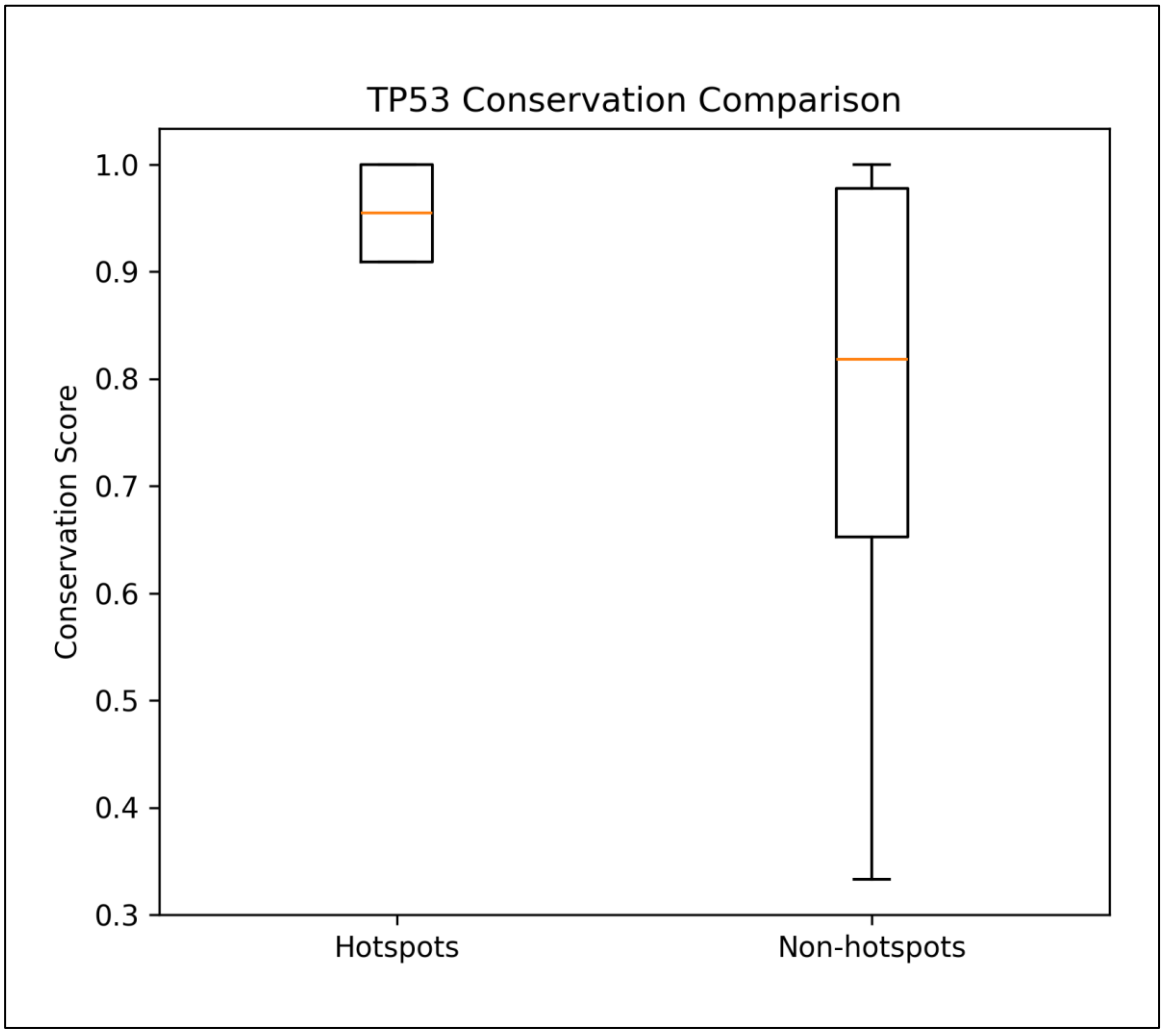

Figure 1. Conservation comparison of hotspot and non-hotspot residues

3.3. Sequence Identity Analysis

Pairwise sequence identity ranged from approximately 69% to 92%, reflecting moderate to high similarity across species and confirming the evolutionary diversity of the dataset.

3.4.Multiple Sequence Alignment Patterns

The alignment revealed strong conservation within the central DNA-binding domain, while terminal regions displayed increased variability.

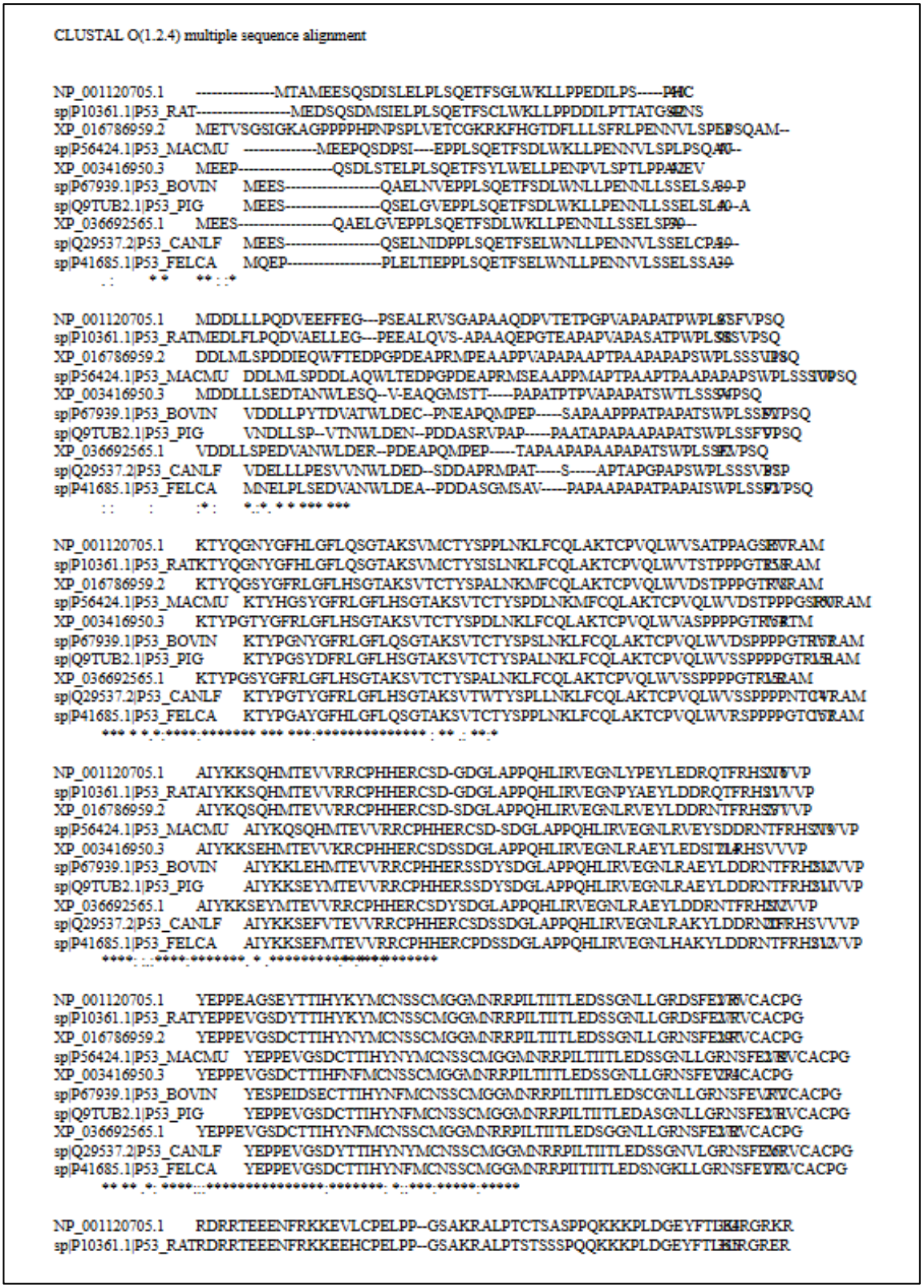

Figure 2. Multiple sequence alignment of TP53

3.5.Phylogenetic Relationships

Phylogenetic analysis demonstrated clustering of TP53 sequences according to species relationships, consistent with evolutionary divergence patterns.

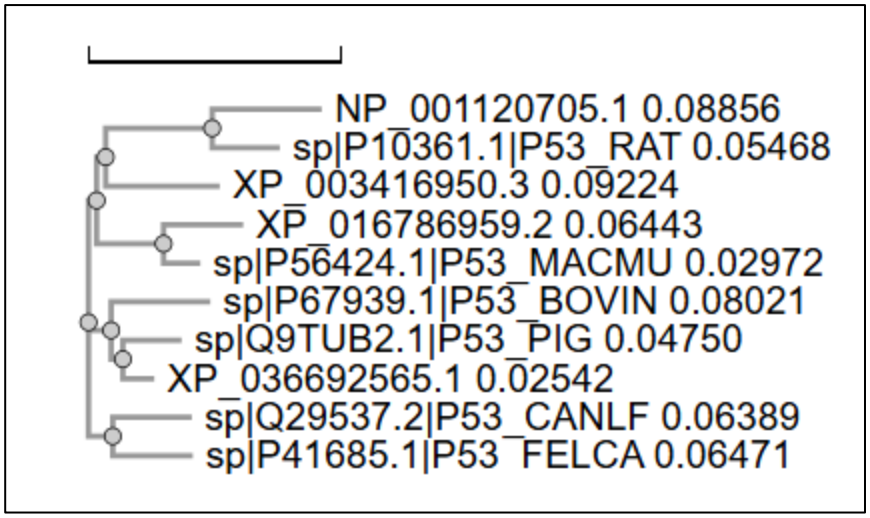

Figure 3. Phylogenetic Tree of TP53

4. DISCUSSION

The present study demonstrates that TP53 mutation hotspots are significantly more conserved than non-hotspot residues across mammalian species. This finding suggests that these positions are subject to strong evolutionary constraint, likely reflecting their essential functional roles.

The localisation of hotspot residues within the DNA-binding domain supports their structural and functional importance [1,2]. Mutations at these positions are known to disrupt protein activity and contribute to cancer development [3].

Evolutionary conservation is typically associated with functional importance, as critical residues are preserved through selective pressure [4]. The observed conservation patterns in this study are consistent with this principle and indicate that hotspot residues are integral to TP53 function.

Furthermore, sequence identity analysis confirms that the dataset captures meaningful evolutionary variation, strengthening the validity of the results. Despite divergence across species, key functional regions remain conserved.

These findings suggest that mutation hotspots in TP53 arise from structural and functional sensitivity rather than random distribution, providing an evolutionary explanation for their recurrent occurrence in cancer.

5. LIMITATIONS

This study is based on computational analysis and does not include experimental validation. Future studies incorporating structural and experimental approaches would provide further insights.

6. CONCLUSION

TP53 mutation hotspots represent evolutionarily constrained residues located within a highly conserved functional domain. Their strong conservation and reduced variability highlight their importance in maintaining protein function and explain their frequent mutation in cancer.

7. AUTHOR CONTRIBUTIONS

Ms Ritika Rajendra. Rawat conceptualised the study, performed sequence analysis, conducted statistical analysis, generated figures, and wrote the manuscript. Mrs Sermarani Nadar contributed to data interpretation and methodology refinement. Dr Gursimran Kaur Uppal contributed to the manuscript review and supervision.

8. FUNDING

No external funding was received.

9. CONFLICT OF INTEREST

The authors declare no conflict of interest.

10. DATA AVAILABILITY

All data used in this study are publicly available from established biological databases. The reference human TP53 protein sequence was obtained from UniProt (accession ID: **P04637**) and NCBI Protein database. Additional TP53 protein sequences from mammalian species were retrieved from NCBI Protein and Ensembl Genome Browser using gene-specific queries.

The accession identifiers for representative sequences used in this study include: **NP_001120705.1** (Mus musculus), **P10361** (Rattus norvegicus), **P56424** (Macaca mulatta), **XP_016786959.2** (Pan troglodytes), **Q29537** (Canis lupus familiaris), **P41685** (Felis catus), **P67939** (Bos taurus), **Q9TUB2** (Sus scrofa), **XP_003416950.3** (Loxodonta africana), and **XP_036692565.1** (Balaenoptera musculus).

The processed multiple sequence alignment (`TP53_MSA.fasta`), conservation analysis results (`TP53_hotspot_analysis.csv`), and computational workflow are provided as supplementary materials. No additional datasets were generated or used beyond publicly accessible resources.

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